Cross-Language Information Retrieval (CLIR) in Global E-Commerce:

Design and Implementation of a Multilingual Semantic Product Retrieval System Using Sentence Transformers

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Abstract

Cross-Language Information Retrieval (CLIR) has emerged as an important research area within Information Retrieval and Natural Language Processing because it enables users to retrieve documents written in one language using queries submitted in another language. As digital platforms increasingly serve international users, multilingual search has become essential for improving accessibility and user experience. Traditional multilingual search systems often depend on machine translation before retrieval, but this approach may introduce translation errors, increase computational cost, and fail to capture domain-specific terminology accurately.

This paper presents the design and implementation of a multilingual semantic search system for an English e-commerce product catalogue. The Amazon Product Dataset obtained from Kaggle was selected as the retrieval corpus, while the multilingual Sentence Transformer model paraphrase-multilingual-MiniLM-L12-v2 was employed to generate semantic embeddings for product titles and multilingual user queries. Product embeddings were generated once and stored for efficient retrieval. User queries written in English, German, Spanish, and Italian were encoded into the same semantic vector space and compared with product embeddings using cosine similarity to retrieve the most relevant products.

Experimental evaluation demonstrates that multilingual sentence embeddings effectively overcome language barriers by retrieving semantically relevant English products without requiring explicit machine translation. The proposed approach provides an efficient and scalable solution for multilingual semantic search and highlights the practical application of transformer-based embedding models within modern e-commerce systems.

Keywords: Cross-Language Information Retrieval, CLIR, Semantic Search, Sentence Transformers, Multilingual Embeddings, Cosine Similarity, Natural Language Processing, Information Retrieval.

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1. Introduction

The rapid expansion of global e-commerce has transformed the way consumers search for products and services. Online marketplaces now serve customers from diverse linguistic and cultural backgrounds, enabling businesses to reach international audiences more effectively than ever before. Despite this global reach, many e-commerce platforms continue to maintain product catalogues in a single language, most commonly English. Consequently, users who submit search queries in their native language often experience difficulties in locating relevant products because traditional search engines primarily rely on lexical or keyword matching rather than understanding the semantic meaning of a query (Manning, Raghavan, & Schütze, 2008).

Conventional information retrieval systems compare the words contained in a user's query with the words indexed in a document or product description. Although this approach is computationally efficient, it performs poorly when the query and the target document are written in different languages or when different terms are used to express the same concept. For example, a Spanish-speaking customer searching for *"auriculares inalámbricos"* expects to find wireless headphones, yet a keyword-based English product catalogue may fail to return relevant results because there is no direct lexical overlap between the query and the indexed product titles. Such limitations negatively affect user experience, product discoverability, and ultimately business performance (Baeza-Yates & Ribeiro-Neto, 2011).

Cross-Language Information Retrieval (CLIR) addresses this challenge by enabling users to submit queries in one language while retrieving relevant documents or products written in another language. Early CLIR systems largely depended on machine translation, translating either the query or the document collection before retrieval. Although effective in some scenarios, translation-based approaches introduce additional computational overhead and may produce inaccurate translations of domain-specific terminology, abbreviations, or product names. These limitations have encouraged researchers to investigate multilingual semantic retrieval methods that represent textual meaning independently of language (Nie, 2010).

Recent advances in Natural Language Processing (NLP), particularly transformer-based language models, have significantly improved multilingual retrieval capabilities. Transformer architectures have demonstrated the ability to capture contextual information more effectively than traditional neural networks by modelling relationships between words through self-attention mechanisms (Vaswani et al., 2017). Building on these advances, Sentence-BERT (SBERT) introduced an efficient framework for generating semantically meaningful sentence embeddings that enable fast similarity comparison while maintaining high retrieval accuracy (Reimers & Gurevych, 2019). Furthermore, multilingual SentenceTransformer models extend this capability across more than fifty languages by mapping semantically similar sentences into a shared embedding space, thereby allowing direct comparison between multilingual queries and English documents without requiring explicit translation (Reimers & Gurevych, 2020).

The objective of this study is to design and implement a Cross-Language Information Retrieval system for multilingual product search in a global e-commerce environment. In accordance with the project requirements, the system uses an English Amazon Product Dataset as the retrieval corpus and applies the paraphrase-multilingual-MiniLM-L12-v2 Sentence Transformer model to generate semantic embeddings for both product titles and multilingual user queries. Retrieval is performed by computing cosine similarity between the embedded query and the embedded product catalogue, allowing semantically related products to be identified even when different languages are used. This implementation reflects the project pipeline specified for the course, which emphasises multilingual semantic search without a separate translation stage.

To evaluate the effectiveness of the proposed approach, multilingual search queries in English, German, Spanish, and Italian are tested against the English product catalogue. The retrieved results are analysed using cosine similarity scores and visualised through similarity heatmaps. This evaluation demonstrates how multilingual sentence embeddings enable language-independent semantic retrieval while reducing the limitations associated with conventional keyword-based search and translation-dependent retrieval systems.

The remainder of this paper is organised as follows. Chapter 2 reviews the theoretical foundations of Information Retrieval, Cross-Language Information Retrieval, multilingual sentence transformers, semantic embeddings, cosine similarity, and related research. Chapter 3 describes the methodology adopted for dataset preparation, preprocessing, embedding generation, and multilingual retrieval. Chapter 4 presents the experimental results together with a discussion of multilingual search performance and similarity analysis. Finally, Chapter 5 summarises the main findings, discusses the limitations of the proposed approach, and outlines potential directions for future research.

2. Theory: State of Research and Academic Debate

The rapid growth of digital information has significantly increased the demand for efficient methods of storing, organising, and retrieving data. Information Retrieval (IR) has therefore become one of the most important research areas in computer science, particularly within Natural Language Processing (NLP), machine learning, and data science. Modern retrieval systems are expected not only to locate documents containing specific keywords but also to understand the semantic meaning of user queries in order to return the most relevant results. Consequently, researchers have progressively shifted from traditional lexical retrieval techniques towards neural and semantic retrieval approaches that better capture contextual relationships between documents and queries (Manning, Raghavan, & Schütze, 2008; Lin, Nogueira, & Yates, 2021).

Cross-Language Information Retrieval (CLIR) represents an important extension of traditional Information Retrieval by enabling users to retrieve documents written in one language using queries submitted in another language. The emergence of multilingual transformer models has fundamentally changed the way cross-language retrieval systems are designed, replacing explicit machine translation pipelines with multilingual semantic embeddings capable of representing multiple languages within a shared vector space (Reimers & Gurevych, 2020). Before discussing these modern approaches, it is necessary to review the theoretical foundations of Information Retrieval and the evolution of retrieval technologies.

2.1 Information Retrieval

Information Retrieval (IR) is the discipline concerned with finding relevant information from large collections of structured or unstructured data in response to a user's information need. Unlike database systems, which retrieve exact records using structured queries, information retrieval systems operate primarily on textual data where relevance is determined by similarity rather than exact matching. The primary objective of an IR system is therefore to maximise the retrieval of relevant documents while minimising the retrieval of irrelevant information (Manning et al., 2008).

Information retrieval systems have become indispensable in numerous real-world applications, including web search engines, digital libraries, recommendation systems, question-answering platforms, legal document retrieval, medical literature search, and e-commerce product discovery. Every time a user submits a search query on platforms such as Google, Amazon, or academic databases, an information retrieval system identifies, ranks, and presents the documents that are considered most relevant to the user's request (Baeza-Yates & Ribeiro-Neto, 2011).

A conventional information retrieval system generally consists of several sequential stages. The first stage involves collecting documents from one or more information sources. These documents undergo preprocessing operations such as text normalisation, tokenisation, stop-word removal, punctuation removal, and stemming or lemmatisation to improve retrieval efficiency. Following preprocessing, the documents are indexed to facilitate rapid searching. When a user submits a query, the same preprocessing operations are applied before the retrieval algorithm compares the processed query with the indexed documents. Finally, the retrieved documents are ranked according to their estimated relevance, and the highest-ranked results are presented to the user (Manning et al., 2008).

Historically, keyword-based retrieval has dominated the field of Information Retrieval. Traditional retrieval models such as the Boolean Retrieval Model, the Vector Space Model (VSM), and the Probabilistic Retrieval Model compare documents and queries primarily through lexical similarity. One of the most widely adopted weighting schemes is Term Frequency–Inverse Document Frequency (TF-IDF), which measures the importance of a word within a document relative to its occurrence across the entire document collection. Terms that occur frequently within a particular document but rarely across the collection receive higher weights because they are considered more informative for distinguishing relevant documents (Salton & Buckley, 1988).

Although lexical retrieval methods remain computationally efficient, they exhibit several important limitations. Most significantly, they rely on exact or near-exact word matching, making them unable to recognise synonyms, paraphrases, contextual relationships, or semantic similarity between different expressions. For example, a user searching for "wireless headphones" may fail to retrieve products described as "Bluetooth earphones", despite the two expressions referring to closely related concepts. Similarly, keyword-based systems generally perform poorly when documents and queries are written in different languages because no lexical overlap exists between the terms being compared (Manning et al., 2008).

The limitations of lexical retrieval motivated the development of semantic retrieval approaches that focus on understanding the meaning of text rather than its exact wording. Advances in deep learning have enabled neural language models to represent words, sentences, and documents as dense numerical vectors that preserve contextual and semantic information. Rather than comparing individual words, neural retrieval systems compare vector representations that encode the overall meaning of textual content. Documents discussing similar concepts therefore appear close together within a high-dimensional embedding space, even when they contain entirely different vocabulary (Devlin et al., 2019).

The introduction of transformer-based architectures has further transformed Information Retrieval by significantly improving contextual language understanding. Transformer models employ self-attention mechanisms that allow each word to be interpreted within the context of surrounding words, thereby producing richer semantic representations than previous recurrent neural network architectures (Vaswani et al., 2017). Building upon these developments, dense retrieval models such as Dense Passage Retrieval (DPR), Sentence-BERT (SBERT), and ColBERT have demonstrated substantial improvements over traditional retrieval systems by learning semantic representations that better capture user intent and document relevance (Karpukhin et al., 2020; Khattab & Zaharia, 2020; Reimers & Gurevych, 2019).

Modern Information Retrieval therefore represents a transition from lexical matching towards semantic understanding. Instead of asking whether two texts contain identical words, contemporary retrieval systems attempt to determine whether they express the same underlying meaning. This shift provides the theoretical foundation for Cross-Language Information Retrieval, where semantic similarity rather than lexical similarity enables effective retrieval across different languages.

2.2 Cross-Language Information Retrieval (CLIR)

Cross-Language Information Retrieval (CLIR) is a specialised branch of Information Retrieval that enables users to submit search queries in one language while retrieving relevant documents written in another language. Unlike traditional monolingual retrieval systems, which assume that both the query and the document collection are written in the same language, CLIR addresses the linguistic barriers that exist in multilingual information systems. As globalisation has increased the availability of multilingual digital content, CLIR has become an important research area in Natural Language Processing (NLP), multilingual information access, and semantic search (Nie, 2010).

The primary objective of CLIR is to overcome language differences without requiring users to manually translate their search queries. This capability is particularly valuable in applications such as international e-commerce, multilingual digital libraries, legal information systems, healthcare databases, and government portals, where information is often stored in one language while users search in another. For example, an online retailer may maintain its product catalogue in English while serving customers who search in German, Spanish, Italian, or French. Without an effective CLIR system, users may fail to retrieve relevant products simply because the query language differs from the language used in the catalogue (Oard & Diekema, 1998).

Early CLIR research primarily relied on translation-based retrieval methods. These systems translated either the user's query into the document language, translated the document collection into the query language, or translated both into a common intermediate language before retrieval. Query translation became the most widely adopted approach because it required translating only a small amount of text and was therefore computationally less expensive than translating an entire document collection. However, the overall performance of translation-based retrieval depended heavily on the quality of the machine translation system. Translation errors frequently resulted in incorrect retrieval, particularly for short search queries that lacked sufficient contextual information (Nie, 2010).

One of the major limitations of machine translation in information retrieval is its inability to consistently preserve domain-specific terminology. Product names, abbreviations, technical specifications, and brand names often have multiple possible translations or should not be translated at all. For example, a product described as "wireless earbuds" may be translated differently depending on the translation model, while branded terms may remain untranslated or be translated incorrectly. Such inconsistencies reduce retrieval accuracy because the translated query no longer accurately reflects the user's original search intent (Ballesteros & Croft, 1998).

Another challenge associated with translation-based CLIR involves query ambiguity. Search queries submitted to e-commerce platforms are typically very short, often containing only one to three words. Because these short queries provide limited linguistic context, machine translation systems frequently struggle to determine the intended meaning. For instance, the English word *"apple"* may refer either to the fruit or the technology company. Without sufficient contextual information, translation systems may produce ambiguous or incorrect translations, ultimately affecting retrieval quality (Oard & Diekema, 1998).

The emergence of multilingual word embeddings represented a significant advancement in Cross-Language Information Retrieval. Instead of explicitly translating text, multilingual embedding models project words from different languages into a shared vector space, where semantically similar words are positioned close together regardless of their original language. This approach enables retrieval systems to compare semantic meaning directly rather than relying on exact lexical correspondence. Early multilingual embedding techniques employed bilingual dictionaries or aligned parallel corpora to construct cross-lingual vector spaces, thereby reducing the dependence on machine translation (Ruder, Vulić, & Søgaard, 2019).

Recent developments in transformer-based language models have further transformed CLIR by enabling multilingual sentence representations that capture semantic information at the sentence level rather than the individual word level. Models such as multilingual BERT (mBERT), XLM-RoBERTa (XLM-R), and multilingual Sentence-BERT learn contextual representations from large multilingual corpora, allowing semantically equivalent sentences from different languages to occupy nearby locations within the same embedding space (Conneau et al., 2020; Devlin et al., 2019). As a result, multilingual retrieval systems no longer require an explicit translation stage, thereby simplifying the retrieval pipeline while improving both efficiency and retrieval effectiveness.

Among these approaches, Sentence Transformers have become particularly popular for semantic retrieval tasks because they generate fixed-length sentence embeddings that can be compared efficiently using similarity measures such as cosine similarity. Unlike the original

BERT architecture, which is computationally expensive for sentence comparison, Sentence-BERT employs a Siamese neural network architecture that produces semantically meaningful embeddings suitable for large-scale retrieval applications (Reimers & Gurevych, 2019). Multilingual versions of Sentence-BERT further extend this capability by aligning sentence embeddings from multiple languages into a common semantic space through knowledge distillation techniques (Reimers & Gurevych, 2020).

The present study adopts this multilingual embedding-based approach rather than a translation-based retrieval strategy. The English Amazon Product Dataset serves as the document collection, while multilingual user queries in German, Spanish, Italian, and English are encoded using the paraphrase-multilingual-MiniLM-L12-v2 model. Because both product titles and user queries are represented within the same embedding space, relevant products can be retrieved based on semantic similarity instead of literal word overlap. This approach eliminates the need for machine translation and enables language-independent retrieval that better preserves the meaning of user queries. The project implementation therefore reflects the modern CLIR pipeline described in the course material, where multilingual queries and English catalogue entries are encoded into a shared semantic space before ranking with cosine similarity.

Current research indicates that multilingual semantic retrieval provides several advantages over traditional translation-based systems. It reduces translation latency, minimises semantic information loss, improves retrieval accuracy for short user queries, and scales more effectively across multiple languages. Nevertheless, challenges remain regarding computational cost, multilingual training data, and retrieval quality for low-resource languages. These issues continue to motivate active research in multilingual representation learning and neural information retrieval, making CLIR one of the fastest-growing research areas within modern Information Retrieval.

2.3 Multilingual Sentence Transformers

The rapid advancement of transformer-based language models has significantly improved the ability of computers to understand natural language. Before the introduction of transformers, Natural Language Processing (NLP) systems primarily relied on recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks. Although these architectures were effective for sequential data, they processed text token by token, making it difficult to capture long-range dependencies and limiting computational efficiency. To overcome these limitations, Vaswani et al. (2017) introduced the Transformer architecture, which relies entirely on a self-attention mechanism instead of recurrence. This innovation enabled models to process complete sentences simultaneously while learning contextual relationships between words more effectively.

The success of the Transformer architecture led to the development of Bidirectional Encoder Representations from Transformers (BERT), introduced by Devlin et al. (2019). Unlike earlier language models, BERT learns bidirectional contextual representations by considering both the left and right context of each word during training. As a result, BERT achieved state-of-the-art performance across numerous Natural Language Processing tasks, including text classification, question answering, named entity recognition, and semantic similarity. Despite its excellent language understanding capabilities, BERT was not originally designed for efficient sentence comparison because obtaining sentence similarity scores required computationally expensive pairwise inference for every query-document combination.

To address this limitation, Reimers and Gurevych (2019) proposed Sentence-BERT (SBERT), a modification of the original BERT architecture that generates fixed-length sentence embeddings using a Siamese neural network structure. Instead of comparing two sentences simultaneously during inference, SBERT independently encodes each sentence into a dense vector representation. These embeddings can then be compared efficiently using similarity measures such as cosine similarity. This significantly reduces computational complexity and makes SBERT particularly suitable for semantic search, information retrieval, clustering, and recommendation systems.

An important extension of Sentence-BERT is the development of multilingual models capable of representing multiple languages within a shared semantic embedding space. Reimers and Gurevych (2020) introduced multilingual sentence embeddings through knowledge distillation, where a multilingual student model learns to reproduce the embeddings generated by a high-performing monolingual teacher model. During training, semantically equivalent sentences written in different languages are mapped to nearby locations within the same vector space. Consequently, a sentence written in German, Spanish, or Italian can be directly compared with an English sentence without requiring explicit translation.

The ability to represent multiple languages within a common embedding space has made multilingual Sentence Transformers particularly valuable for Cross-Language Information Retrieval. Rather than translating user queries into the language of the document collection, multilingual embeddings allow documents and queries to be compared directly based on semantic similarity. This approach reduces translation errors, decreases computational overhead, and improves retrieval performance for multilingual search applications (Reimers & Gurevych, 2020).

For the implementation presented in this study, the paraphrase-multilingual-MiniLM-L12-v2 model available through the Sentence Transformers library was selected. This model supports more than fifty languages while maintaining a relatively small model size compared with larger transformer architectures. According to the project requirements, it provides an appropriate balance between computational efficiency and semantic retrieval quality, making it suitable for deployment in environments with limited computational resources while still achieving strong multilingual performance.

The selected model encodes both English product titles and multilingual user queries into 384-dimensional dense vector representations. Because every sentence is projected into the same semantic embedding space, queries submitted in German, Spanish, Italian, or English can retrieve relevant English product titles without requiring a separate translation component. This shared semantic representation forms the core of the multilingual retrieval pipeline implemented in this project and demonstrates how transformer-based embedding models enable language-independent semantic search in modern e-commerce systems.

2.4 Semantic Embeddings

After textual information has been transformed into dense vector representations, an appropriate similarity measure is required to determine how closely two vectors are related. Cosine similarity is one of the most widely used similarity metrics in Information Retrieval because it measures the angle between two vectors rather than their magnitude. Documents with similar semantic content produce vectors pointing in nearly the same direction, resulting in cosine similarity values close to one (Manning et al., 2008).

Mathematically, cosine similarity is calculated as the cosine of the angle between two vectors.

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\| A \| \| B \|}$$

The resulting similarity score ranges from −1 to 1, where:

- 1 indicates identical semantic direction.
- 0 indicates no semantic relationship.
- −1 indicates opposite semantic meaning.

In most sentence embedding applications, similarity values fall between 0 and 1, as embedding vectors generally represent semantically meaningful textual information. Higher cosine similarity scores indicate stronger semantic relationships between the user query and candidate documents.

Cosine similarity offers several advantages for semantic retrieval. It is computationally efficient, independent of vector magnitude, and highly suitable for comparing dense sentence embeddings generated by Transformer-based models. These characteristics have made cosine similarity the standard similarity metric in many modern semantic retrieval systems, including Sentence Transformers (Reimers & Gurevych, 2019).

Within this project, cosine similarity is calculated between the embedding generated for each multilingual query and every embedding in the English product catalogue. Products are subsequently ranked according to their similarity scores, and the highest-ranked products are returned as the retrieval results. This ranking mechanism enables semantically related products to be identified even when the query and product descriptions are written in different languages. The implementation follows the retrieval pipeline specified for the course project, where multilingual embeddings are compared using cosine similarity to produce ranked search results.

2.6 Related Work

Cross-Language Information Retrieval has been an active research area for more than two decades. Early research primarily focused on dictionary-based and machine translation-based retrieval methods. These approaches demonstrated that multilingual retrieval was feasible but also highlighted limitations related to translation ambiguity, vocabulary mismatch, and computational overhead (Oard & Diekema, 1998; Nie, 2010).

The emergence of distributed word representations significantly improved multilingual retrieval by allowing semantically related words from different languages to be represented within shared embedding spaces. Cross-lingual embedding techniques reduced dependence on explicit translation while improving retrieval performance across multilingual document collections (Ruder, Vulić, & Søgaard, 2019).

More recently, Transformer-based language models have become the dominant approach for semantic retrieval. BERT introduced contextual language representations that substantially improved many Natural Language Processing tasks (Devlin et al., 2019). Sentence-BERT extended these capabilities by generating efficient sentence embeddings suitable for large-scale retrieval applications (Reimers & Gurevych, 2019). Furthermore, multilingual Sentence Transformers enabled semantic comparison across dozens of languages through multilingual knowledge distillation (Reimers & Gurevych, 2020).

Recent studies have demonstrated that dense neural retrieval models consistently outperform traditional lexical retrieval systems in multilingual search tasks because they capture semantic relationships beyond exact word matching (Lin, Nogueira, & Yates, 2021). These advances have encouraged the adoption of semantic retrieval techniques within digital libraries, question-answering systems, recommendation engines, and global e-commerce platforms.

The present study builds upon this body of research by implementing a lightweight multilingual semantic retrieval system using the paraphrase-multilingual-MiniLM-L12-v2 model. Unlike translation-based approaches, the proposed system directly compares multilingual user queries with English product titles through shared sentence embeddings and cosine similarity ranking. This implementation demonstrates how recent developments in multilingual transformer models can be applied to improve product discovery within global e-commerce environments while maintaining computational efficiency.

3. Methodology

3.1 Research Design

This study follows an applied experimental research design to demonstrate the implementation of a Cross-Language Information Retrieval (CLIR) system for multilingual product search in a global e-commerce environment. Rather than developing a new retrieval algorithm, the project applies existing multilingual Natural Language Processing (NLP) techniques to build a functional prototype capable of retrieving English product information using multilingual search queries.

The implementation was developed using Python in the Google Colab environment. Google Colab was selected because it provides a cloud-based platform with pre-configured Python libraries, eliminating the need for local software installation while supporting efficient experimentation with transformer-based language models. The project focuses on semantic retrieval instead of traditional keyword matching by using multilingual sentence embeddings generated through the Sentence Transformers framework.

The overall objective of the implementation is to evaluate whether multilingual sentence embeddings can accurately retrieve English product titles when users submit search queries in different languages, including German, Spanish, Italian, and English.

3.2 Dataset Preparation

The retrieval corpus used in this study consists of an English product catalogue based on an Amazon-style e-commerce dataset. Each product record contains a product title together with its associated category and descriptive information. Since the objective of the project is to evaluate multilingual retrieval, the product catalogue

3.3 Model Selection

The implementation employs the paraphrase-multilingual-MiniLM-L12-v2 model from the Sentence Transformers library. This model was selected because it provides multilingual sentence embeddings while maintaining relatively low computational requirements. According to the model documentation, it supports more than fifty languages, making it suitable for multilingual semantic retrieval tasks.

Compared with larger transformer architectures, MiniLM offers faster inference while maintaining strong semantic representation quality. Since the project focuses on demonstrating multilingual retrieval rather than maximising benchmark performance, this model provides an appropriate balance between efficiency and retrieval accuracy.

The Sentence Transformer framework converts each sentence into a fixed-length dense embedding. These embeddings represent the semantic meaning of the complete sentence rather than individual words, allowing semantically similar sentences written in different languages to occupy nearby locations within a shared vector space. Consequently, multilingual queries can be compared directly with English product titles without requiring an intermediate machine translation stage.

3.4 Retrieval Pipeline

The multilingual retrieval pipeline consists of several sequential stages.

Step 1: Dataset Loading

The English product catalogue is imported into the Google Colab environment using the Pandas library. Product titles are extracted to form the retrieval corpus.

Step 2: Text Preprocessing

Basic preprocessing is performed to normalise textual data and prepare product titles for embedding generation.

Step 3: Model Initialisation

The **paraphrase-multilingual-MiniLM-L12-v2** Sentence Transformer model is loaded from the Hugging Face repository.

Step 4: Product Embedding Generation

Each product title is encoded into a dense numerical vector using the selected Sentence Transformer model. These embeddings are computed once and stored for later comparison.

Step 5: Multilingual Query Encoding

User queries written in English, German, Spanish, or Italian are encoded using the same Sentence Transformer model. Because both queries and products are represented within the same embedding space, they become directly comparable regardless of language.

Step 6: Similarity Computation

Cosine similarity is calculated between the query embedding and every product embedding within the catalogue. Higher similarity values indicate stronger semantic relationships.

Step 7: Ranking

Products are ranked according to their cosine similarity scores. The highest-ranked products are returned as the retrieval results presented to the user.

This pipeline demonstrates multilingual semantic retrieval without relying on explicit machine translation, illustrating the effectiveness of multilingual sentence embeddings for modern e-commerce search applications.

3.5 Evaluation Method

The proposed CLIR system was evaluated using multilingual search queries representing realistic product searches in English, German, Spanish, and Italian. For each query, cosine similarity scores were calculated against every product within the English product catalogue.

Retrieval performance was assessed by examining whether semantically relevant products appeared among the highest-ranked search results. In addition to ranked retrieval lists, cosine similarity matrices were visualised using heatmaps to illustrate the semantic relationships between multilingual queries and English product titles. These visualisations provide an intuitive representation of retrieval performance by highlighting stronger and weaker semantic matches across different languages.

The evaluation focuses primarily on demonstrating the effectiveness of multilingual semantic retrieval rather than conducting large-scale benchmarking. Consequently, the project emphasises qualitative analysis of retrieval behaviour together with similarity score interpretation.

4. Analysis and Discussion

4.1 Experimental Setup

The objective of the experimental evaluation was to assess the effectiveness of the proposed Cross-Language Information Retrieval (CLIR) system in retrieving English product titles from multilingual user queries. The implementation was executed in the Google Collab environment using Python and several open-source libraries, including Pandas for data processing, Sentence Transformers for embedding generation, Scikit-learn for cosine similarity computation, Matplotlib for visualisation, and NumPy for numerical operations.

The retrieval corpus consisted of an English product catalogue containing representative product titles from multiple categories. User queries were prepared in English, German, Spanish, and Italian to simulate realistic multilingual search behaviour within an international e-commerce platform. Both product titles and user queries were encoded using the paraphrase-multilingual-MiniLM-L12-v2 model, producing dense sentence embeddings within a shared semantic vector space. Cosine similarity scores were then calculated between each query embedding and every product embedding to determine semantic relevance. Products with the highest similarity scores were returned as the final retrieval results.

The experimental workflow followed the same sequence described in the methodology chapter:

1. Load the English product catalogue
2. Generate embeddings for all product titles.
3. Encode multilingual user queries.
4. Calculate cosine similarity.
5. Rank products according to similarity scores.
6. Visualise similarity scores using heatmaps.

This evaluation demonstrates the complete multilingual semantic retrieval pipeline without relying on machine translation.

4.2 Retrieval Results

The experimental results demonstrate that multilingual sentence embeddings successfully identify semantically related products even when the query language differs from the language used in the product catalogue. Instead of depending on exact keyword overlap, the retrieval system compares semantic representations learned during multilingual transformer training.

English queries consistently returned the correct products with the highest cosine similarity scores because both the query and the catalogue were written in the same language. More importantly, German, Spanish, and Italian queries also retrieved the corresponding English product titles despite having completely different vocabularies. This indicates that the multilingual Sentence Transformer effectively projected semantically equivalent sentences into a common embedding space.

For example, a German query referring to running shoes successfully matched the English product "Premium Leather Running Shoes." Similarly, Spanish and Italian queries describing wireless headphones retrieved the corresponding English product with high similarity values. These results illustrate that semantic similarity rather than lexical similarity determines document relevance within the proposed retrieval framework.

The ranking process also demonstrated that relevant products consistently appeared among the highest-ranked search results. Although similarity scores varied slightly across different languages, the correct products remained near the top of the ranked list, indicating that the multilingual embedding model preserved semantic meaning effectively across language boundaries.

These observations support previous research showing that multilingual sentence embeddings substantially improve retrieval performance compared with traditional keyword-based retrieval systems, particularly for multilingual search applications (Reimers & Gurevych, 2020).

4.3 Cosine Similarity Analysis

Cosine similarity served as the primary ranking metric throughout the retrieval process. Higher similarity scores indicated stronger semantic relationships between multilingual user queries and English product titles.

The similarity matrices generated during experimentation were visualised using heatmaps. Each row represented a multilingual user query, while each column represented an English product title. Darker cells indicated higher cosine similarity values and therefore stronger semantic correspondence between the query and the product. The heatmaps provided an intuitive representation of how multilingual embeddings clustered semantically related products despite language differences.

Several important observations can be made from these visualisations.

First, the highest similarity values were concentrated around the correct query–product pairs, indicating that the multilingual embedding model successfully captured semantic relationships. Second, unrelated products generally produced much lower similarity values, demonstrating that the embedding space effectively separated different semantic concepts. Finally, small similarity values observed between related product categories reflected the ability of the model to recognise broader semantic relationships while still assigning the highest scores to the most relevant products.

Overall, the cosine similarity analysis confirms that semantic embeddings provide a meaningful representation of multilingual textual information and can effectively support cross-language product retrieval.

4.4 Discussion

The experimental findings demonstrate several advantages of multilingual semantic retrieval over conventional keyword-based search systems.

The most significant advantage is the elimination of explicit machine translation. Traditional CLIR systems translate user queries before retrieval, introducing additional processing time and the possibility of translation errors. In contrast, the proposed system compares multilingual queries and English product titles directly within a shared embedding space, simplifying the retrieval pipeline while preserving semantic meaning.

Another advantage is improved robustness against vocabulary differences. Keyword-based search engines frequently fail when users employ synonyms, abbreviations, or different linguistic expressions. Multilingual sentence embeddings overcome this limitation by focusing on sentence meaning rather than exact word overlap. Consequently, users can retrieve relevant products even when different languages or alternative expressions are used.

The lightweight paraphrase-multilingual-MiniLM-L12-v2 model also proved appropriate for this implementation because it combines multilingual support with relatively fast inference. This makes it suitable for educational prototypes and small to medium-sized semantic retrieval applications where computational resources may be limited.

Despite these positive findings, several limitations should be acknowledged. The experimental evaluation used a relatively small product catalogue and a manually prepared set of multilingual queries. Consequently, the results cannot be directly generalised to large commercial e-commerce platforms containing millions of products. Furthermore, the evaluation focused primarily on qualitative retrieval behaviour rather than quantitative benchmark metrics such as Precision@K, Recall@K, or Mean Reciprocal Rank (MRR).

Another limitation is that the retrieval system relies solely on semantic similarity between product titles and user queries. Additional product metadata such as descriptions, specifications, customer reviews, popularity signals, and user preferences were not incorporated into the ranking process. Including these additional information sources could further improve retrieval effectiveness.

Nevertheless, the implementation successfully demonstrates the practical application of multilingual sentence embeddings within Cross-Language Information Retrieval. The results support existing literature indicating that transformer-based semantic retrieval systems provide a powerful alternative to traditional translation-dependent retrieval approaches, particularly in multilingual e-commerce environments.

4.5 Chapter Summary

This chapter evaluated the proposed multilingual semantic retrieval system using an English product catalogue and multilingual user queries. The experimental results demonstrated that the Sentence Transformer model successfully retrieved semantically relevant English products from German, Spanish, Italian, and English search queries. Cosine similarity analysis and heatmap visualisation further illustrated that semantically related queries and products occupied nearby locations within the shared embedding space.

Although the evaluation was conducted on a relatively small dataset, the implementation confirmed the feasibility of multilingual semantic retrieval without explicit machine translation. The findings indicate that transformer-based multilingual embeddings can significantly improve product discovery across language boundaries while maintaining computational efficiency, making them a promising solution for modern global e-commerce search systems.

5. Conclusion

5.1 Summary of the Study

The increasing globalisation of e-commerce has created a growing demand for information retrieval systems that can effectively process multilingual user queries. Traditional keyword-based search systems often struggle to retrieve relevant information when users search in different languages because they depend heavily on exact lexical matching. Cross-Language Information Retrieval (CLIR) addresses this challenge by enabling users to search for information in one language while retrieving documents written in another.

This seminar project presented the design and implementation of a multilingual semantic retrieval system using modern transformer-based language models. The system was developed using Python in the Google Colab environment and employed the paraphrase-multilingual-MiniLM-L12-v2 Sentence Transformer model to generate multilingual sentence embeddings. Rather than relying on machine translation, both English product titles and multilingual user queries were mapped into a shared semantic embedding space, allowing direct comparison through cosine similarity.

The implementation demonstrated the complete retrieval pipeline, including dataset preparation, text preprocessing, embedding generation, multilingual query encoding, similarity computation, and ranking of search results. An English product catalogue served as the retrieval corpus, while user queries were prepared in English, German, Spanish, and Italian to evaluate multilingual retrieval performance. Experimental results showed that the proposed approach successfully retrieved semantically relevant English product titles even when the queries were written in different languages. The similarity heatmaps further illustrated that multilingual sentence embeddings effectively captured semantic relationships across languages.

Overall, the study demonstrates that multilingual transformer models provide an efficient and practical solution for Cross-Language Information Retrieval. By focusing on semantic meaning instead of exact keyword matching, the proposed system improves multilingual search capability and highlights the potential of transformer-based retrieval methods for modern e-commerce applications.

5.2 Limitations

Although the proposed CLIR system achieved its objective of demonstrating multilingual semantic retrieval, several limitations should be acknowledged.

First, the evaluation was conducted using a relatively small English product catalogue and a limited number of multilingual queries. While this dataset was sufficient to demonstrate the retrieval pipeline, it does not represent the scale or diversity of commercial e-commerce platforms containing millions of products.

Second, the experimental evaluation primarily focused on qualitative analysis of retrieval behaviour. The effectiveness of the system was assessed through ranked retrieval results, cosine similarity values, and similarity heatmaps rather than comprehensive quantitative evaluation metrics such as Precision@K, Recall@K, Mean Average Precision (MAP), or Mean Reciprocal Rank (MRR). Future evaluations using standardised benchmark datasets would provide a more detailed assessment of retrieval performance.

Another limitation is that the retrieval model considered only product titles when generating semantic embeddings. Additional information such as product descriptions, specifications, customer reviews, popularity indicators, and user interaction history was not incorporated into the retrieval process. Including these features could improve ranking quality and overall search performance.

Finally, the study evaluated a single multilingual Sentence Transformer model. Although the selected model provides an effective balance between computational efficiency and retrieval quality, alternative multilingual transformer architectures may offer improved performance for specific retrieval tasks.

5.3 Future Work

Several opportunities exist to extend and improve the proposed multilingual retrieval system.

One possible direction is the use of larger multilingual product datasets containing products from multiple categories and languages. Evaluating the retrieval system on more diverse datasets would provide stronger evidence of its scalability and practical applicability.

Future research could also compare different multilingual embedding models, such as multilingual BERT, XLM-RoBERTa, and other Sentence Transformer variants, to identify the most suitable architecture for multilingual e-commerce search. Such comparisons would provide valuable insights into the trade-offs between retrieval accuracy, computational efficiency, and inference speed.

Another promising extension is the integration of vector databases such as FAISS or other approximate nearest neighbour search techniques. These technologies would significantly improve retrieval speed when processing very large product catalogues and would make real-time semantic search more practical for commercial applications.

The retrieval system could also be enhanced by incorporating additional product metadata, customer reviews, user preferences, and behavioural information into the ranking process. Combining semantic similarity with traditional ranking signals may further improve retrieval quality and user satisfaction.

Finally, multilingual semantic retrieval could be integrated into recommendation systems, conversational shopping assistants, and intelligent e-commerce platforms. Such applications would allow users from different linguistic backgrounds to interact naturally with online product catalogues without requiring explicit translation, thereby improving accessibility and the overall user experience.

5.4 Concluding Remarks

This seminar project demonstrated the successful implementation of a multilingual semantic retrieval system for Cross-Language Information Retrieval using transformer-based sentence embeddings. The results indicate that multilingual Sentence Transformers can effectively bridge language barriers by representing queries and documents within a shared semantic space. The implementation showed that relevant English product information can be retrieved accurately from multilingual user queries without relying on machine translation.

As multilingual digital platforms continue to expand, semantic retrieval techniques are expected to play an increasingly important role in improving information accessibility and user experience. The proposed implementation provides a practical example of how recent advances in Natural Language Processing can be applied to multilingual information retrieval and serves as a foundation for future research and development in intelligent e-commerce search systems.

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